Enterprise Network Lab: Running Command Manual

Palo Alto PAN-OS • VyOS Enterprise Router • GNS3 Architecture • VPCS Clients

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Time Window: 12:00 PM – 01:50 PM	**VyOS Version:** 1.5-rolling (Kea DHCP)	**Firewall:** PA-VM PAN-OS 10.x

1. Host Performance & Memory Optimization Commands

Executed on Linux host to resolve GNS3 kernel out-of-memory freeze issues by expanding virtual swap memory from 2 GB to 16 GB.

```
# 1. Inspect initial RAM & Swap footprint
free -h
vmstat -s | head -20
ps aux --sort=-%mem | head -15

# 2. Safely deactivate legacy swap file
sudo swapoff /swapfile

# 3. Allocate 16 GB high-speed swap space
sudo fallocate -l 16G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile

# 4. Verify upgraded swap capacity (Total Virtual Memory: ~31 GB)
swapon --show
free -h
```

2. GNS3 Docker Backend & REST API Management

Used for container daemon orchestration, project state queries, and topology link manipulation.

```
# 1. Check Docker container state and restart GNS3 backend
docker ps
docker compose up -d
docker logs --tail 25 gns3-server

# 2. Query GNS3 REST API for active projects and node inventory
curl -s http://127.0.0.1:3080/v2/version
curl -s http://127.0.0.1:3080/v2/projects
curl -s -X POST http://127.0.0.1:3080/v2/projects/700ebbc6-f5cb-4230-b2cf-2b0f57d94006/open

# 3. Disconnect erroneous links via REST API (Firewall-to-Switch cables)
# Delete PA-VM-1 ethernet1/0 <-> Switch1 e0
curl -X DELETE http://127.0.0.1:3080/v2/projects/700ebbc6-f5cb-4230-b2cf-2b0f57d94006/links/1ce3530d-2c8c-4936-a2d3

# Delete PA-VM-1 ethernet1/1 <-> Switch2 e0
curl -X DELETE http://127.0.0.1:3080/v2/projects/700ebbc6-f5cb-4230-b2cf-2b0f57d94006/links/8fa87283-4a88-4bfb-9fff

# Delete Router-to-Firewall temporary cable (eth0 <-> ethernet1/3)
curl -X DELETE http://127.0.0.1:3080/v2/projects/700ebbc6-f5cb-4230-b2cf-2b0f57d94006/links/42e5e1f2-6fa1-4094-8975
```

3. Console Access & Port Mapping Reference

Direct local telnet console endpoints bypassing web browser key interception (enables native Tab completion):

Device Node	Role	Telnet Port	Credentials	Management / Access
PA-VM-1	Next-Gen Firewall	127.0.0.1:5000	admin / Nishu@123@	Web GUI: https://172.17.0.50
VyOS Router	Enterprise Router	127.0.0.1:5022	vyos / vyos	Full CLI Routing & DHCP
PC1	VPCS Client (LAN 1)	127.0.0.1:5002	No Auth	DHCP: 192.168.10.10/24
PC2	VPCS Client (LAN 2)	127.0.0.1:5004	No Auth	DHCP: 192.168.20.12/24
PC8	VPCS Client (LAN 1)	127.0.0.1:5018	No Auth	Cleaned VPCS instance

4. Palo Alto Networks (PAN-OS) CLI Commands

Operational status diagnostics and dependency-ordered configuration deletion.

```
# 1. Operational Mode Status Commands
admin@PA-VM> show interface management       # Verify MGT IP (172.17.0.50)
admin@PA-VM> show interface hardware         # View physical interfaces
admin@PA-VM> show interface all              # View logical interfaces
admin@PA-VM> set cli scripting-mode off       # Enable interactive CLI
admin@PA-VM> set cli pager on                # Enable terminal pager

# 2. Configuration Reset (Dependency Order - Avoids Server Ref Errors)
admin@PA-VM> configure

# Step A: Delete Security & NAT Policies
delete rulebase security rules Allow-InterLAN
delete rulebase security rules Allow-Outbound-Internet.
delete rulebase security rules Allow-Inbound-From-Laptop
delete rulebase nat rules Outbound-NAT

# Step B: Delete Virtual Router Static Routes & Interface Attachments
delete network virtual-router default routing-table ip static-route Default-Gateway
delete network virtual-router default interface

# Step C: Delete Security Zones
delete zone Trust-L1
delete zone Trust-L2
delete zone Untrust

# Step D: Delete Hardware Interface Configurations & Imports
delete import network interface
delete network interface ethernet ethernet1/1
delete network interface ethernet ethernet1/2
delete network interface ethernet ethernet1/3

# Step E: Commit Clean Configuration
commit
run show jobs id 22                          # Monitor commit job progress (FIN OK)
exit
```

5. VyOS Enterprise Router Commands

Gateway interface provisioning, static routing, and Kea dual-subnet DHCP server deployment.

```
# 1. Log in to VyOS Console (vyos / vyos)
configure

# 2. Clean Existing Interface Configurations
delete interfaces ethernet eth0 address
delete interfaces ethernet eth1 address
delete interfaces ethernet eth2 address

# 3. Configure Enterprise Layer 3 Gateways
set interfaces ethernet eth0 address 10.0.0.2/30       # Uplink to Palo Alto
set interfaces ethernet eth1 address 192.168.10.1/24   # Gateway for Switch1 (LAN 1)
set interfaces ethernet eth2 address 192.168.20.1/24   # Gateway for Switch2 (LAN 2)

# 4. Default Gateway pointing to Palo Alto Firewall
set protocols static route 0.0.0.0/0 next-hop 10.0.0.1

# 5. Kea DHCP Server Pool 1 (Switch1 / LAN-1: 192.168.10.0/24)
set service dhcp-server shared-network-name LAN-POOL subnet 192.168.10.0/24 subnet-id 1
set service dhcp-server shared-network-name LAN-POOL subnet 192.168.10.0/24 option default-router 192.168.10.1
set service dhcp-server shared-network-name LAN-POOL subnet 192.168.10.0/24 option name-server 8.8.8.8
set service dhcp-server shared-network-name LAN-POOL subnet 192.168.10.0/24 range 0 start 192.168.10.10
set service dhcp-server shared-network-name LAN-POOL subnet 192.168.10.0/24 range 0 stop 192.168.10.50

# 6. Kea DHCP Server Pool 2 (Switch2 / LAN-2: 192.168.20.0/24)
set service dhcp-server shared-network-name LAN2-POOL subnet 192.168.20.0/24 subnet-id 2
set service dhcp-server shared-network-name LAN2-POOL subnet 192.168.20.0/24 option default-router 192.168.20.1
set service dhcp-server shared-network-name LAN2-POOL subnet 192.168.20.0/24 option name-server 8.8.8.8
set service dhcp-server shared-network-name LAN2-POOL subnet 192.168.20.0/24 range 0 start 192.168.20.10
set service dhcp-server shared-network-name LAN2-POOL subnet 192.168.20.0/24 range 0 stop 192.168.20.50

# 7. Commit and Persist Configuration to Flash
commit
save
exit

# 8. Operational Verification Commands
show interfaces
show service dhcp-server
```

6. VPCS Client Commands & Traffic Verification

Commands executed on end-host VPCS stations (PC1 on Switch1 and PC2 on Switch2):

```
# --- ON PC1 (Switch1 / LAN-1) ---
PC1> clear ip                              # Flush stale configuration
PC1> dhcp                                  # Discover & lease IP from VyOS
# Output: DORA IP 192.168.10.10/24 GW 192.168.10.1
PC1> show ip                               # Verify IP, Mask, Gateway, DNS
PC1> ping 192.168.10.1                     # Ping local VyOS gateway (0.37 ms)

# --- ON PC2 (Switch2 / LAN-2) ---
PC2> clear ip                              # Flush stale configuration
PC2> dhcp                                  # Discover & lease IP from VyOS
# Output: DORA IP 192.168.20.12/24 GW 192.168.20.1
PC2> show ip                               # Verify IP, Mask, Gateway, DNS
PC2> ping 192.168.20.1                     # Ping local VyOS gateway (0.30 ms)

# --- INTER-SWITCH / INTER-VLAN ROUTING VALIDATION ---
PC2> ping 192.168.10.10                    # Ping across VyOS to PC1 on Switch1
# 84 bytes from 192.168.10.10 icmp_seq=1 ttl=63 time=3.409 ms
# 84 bytes from 192.168.10.10 icmp_seq=2 ttl=63 time=0.439 ms (SUCCESS)
```

7. Planned Next Steps: Palo Alto Point-to-Point Uplink

Commands to configure Palo Alto ethernet1/1 and route return traffic back to LAN 1 & LAN 2:

```
admin@PA-VM> configure

# 1. Create Ping Management Profile
set network profiles interface-management-profile allow-ping ping yes

# 2. Configure Point-to-Point Uplink Interface (ethernet1/1)
set network interface ethernet ethernet1/1 layer3 ip 10.0.0.1/30
set network interface ethernet ethernet1/1 layer3 interface-management-profile allow-ping
set network virtual-router default interface ethernet1/1
set zone Router-Zone network layer3 ethernet1/1

# 3. Add Return Routes for Internal LANs via VyOS (10.0.0.2)
set network virtual-router default routing-table ip static-route LAN1-Route destination 192.168.10.0/24 interface e
set network virtual-router default routing-table ip static-route LAN2-Route destination 192.168.20.0/24 interface e

# 4. Security Policy & Commit
set rulebase security rules Allow-All to any from any source any destination any application any service any actior
commit
```